

Xiaomi's Modular Optical System is markedly different

One of the key advancements in Xiaomi's approach is its sensor. The QX100 used a 1-inch sensor, which was impressive for its time but ultimately constrained by slow data processing and lack of direct integration. Xiaomi's new module, however, features a 100-megapixel Micro Four Thirds sensor, a significant leap that rivals professional compact cameras. Combined with a 35mm f/1.4 lens and variable aperture control, Xiaomi's module offers a level of control that was absent in Sony's previous attempt.

Beyond sensor size, Xiaomi's design considerations also differ. The QX100 was a bulky add-on that felt awkward when attached to a phone. Xiaomi's module, while still protruding, is designed with a more compact form factor and a lightweight 6G aspherical lens structure to maintain balance. Early hands-on reports note negligible lag, which we are sure the brand will iron out soon, and the ability to switch between the module and built-in lenses seamlessly. This cohesion suggests a user experience closer to traditional camera systems than previous modular attempts.

Learning from Sony's mistakes

So why did Sony's QX series fail, and why might Xiaomi's approach succeed? The biggest drawback of Sony's implementation was the lag and inconvenience. The camera took seconds to connect, and even when it did, latency in live view made real-time composition difficult. Addition-



ally, reliance on Wi-Fi meant that battery drain was a persistent issue. Xiaomi's module, on the other hand, directly integrates with the phone's hardware, eliminating the latency and power inefficiencies that plagued Sony's implementation.

Another key issue with Sony's approach was its lack of intuitive control. While the QX series had physical buttons, most operations required the app, creating a disjointed

user experience. Xiaomi's module addresses this with physical focus and zoom controls, ensuring that users can interact with the module without always relying on software adjustments.

The one drawback: Compatibility

However, Xiaomi's system is not without its drawbacks. One of the main advantages of the QX10 and QX100 was their universal compatibility – users could attach them to virtually any smartphone. Xiaomi's module, in contrast, is limited to specific Xiaomi models, restricting its potential adoption. While this ensures deep integration, it also limits market penetration and may deter potential users who don't own Xiaomi phones.

Modular Optical System is a step toward modular smartphone photography

Xiaomi's modular camera module represents a well-calculated evolution of a concept that has historically struggled to gain traction. By addressing the connectivity issues, sensor limitations, and user experience pitfalls that plagued Sony's QX series, Xiaomi stands a much better chance at success. However,

its limited compatibility remains a potential hurdle.

If Xiaomi can expand support to a broader range of devices or introduce a universally adaptable variant, this could mark a significant turning point in smartphone photography. Whether this is the future of mobile imaging or another ambitious experiment remains to be seen, but for now, Xiaomi's take on modular optics is a promising step forward. ■